Special Relativity Notes

Transformation of Force and of the Electric Field

6/11/26

Velocities and symbols used throughout (please read first)

- S — the original frame; S' — a frame moving along the $+x$ axis with velocity $\vec{v}$ **relative to** S (a boost along x). $v = |\vec{v}|$.
- $\beta = \dfrac{v}{c}$, $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$ (c = speed of light). These always refer to the **frame** velocity v, never to the charge's velocity.
- $\vec{u} = (u_x, u_y, u_z)$ — velocity of the **test charge** as measured in S.
- $\vec{u}\,' = (u'_x, u'_y, u'_z)$ — velocity of the same test charge as measured in S'.
- $\vec{F}$, $\vec{F}\,'$ — the force on the test charge in S and in S'.
- Unprimed quantities are measured in S; primed quantities in S'.

1 Summary: How Does Force Transform under a Lorentz Transformation?

Forward ($S \to S'$)

$$F'_x = \frac{F_x - \dfrac{\beta}{c}(\vec{F}\cdot\vec{u})}{1 - \dfrac{\beta}{c}u_x}$$

$$F'_y = \frac{F_y}{\gamma\left(1 - \dfrac{\beta\, u_x}{c}\right)}$$

$$F'_z = \frac{F_z}{\gamma\left(1 - \dfrac{\beta\, u_x}{c}\right)}$$

Inverse Lorentz transformation ($S' \to S$)

$$F_x = \frac{F'_x + \dfrac{\beta}{c}(\vec{F}\,'\cdot\vec{u}\,')}{1 + \dfrac{\beta}{c}u'_x}$$

$$F_y = \frac{F'_y}{\gamma\left(1 + \dfrac{\beta\, u'_x}{c}\right)}$$

$$F_z = \frac{F'_z}{\gamma\left(1 + \dfrac{\beta\, u'_x}{c}\right)}$$

What each term means

- F_x, F_y, F_z and F'_x, F'_y, F'_z — components of the force on the test charge in S and in S' (x is the direction of the boost).
- $\vec{F}\cdot\vec{u} = F_x u_x + F_y u_y + F_z u_z$ — the power delivered to the charge in S (its rate of change of energy); $\vec{F}\,'\cdot\vec{u}\,'$ is the same quantity in S'.
- u_x — x-component of the test charge's velocity in S; u'_x — the same in S'.
- $\beta/c = v/c^2$; $\gamma = 1/\sqrt{1 - v^2/c^2}$; v is the velocity of S' relative to S.
- The **inverse** transformation is obtained from the forward one by swapping primed and unprimed quantities and replacing $v \to -v$ (so $\beta \to -\beta$, which flips the signs; γ is

unchanged because it depends on v^2).

2 The Lorentz Force

If a charged particle moves in an electromagnetic field, the force experienced by it is called the **Lorentz force**:

$$\vec{F} = q\left(\vec{E} + \vec{u} \times \vec{B}\right)$$

where: q = charge of the particle (a Lorentz invariant, the same in every frame); $\vec{E}$ = electric field and $\vec{B}$ = magnetic field, both measured in the frame being used; $\vec{u}$ = velocity of the charged particle in that frame; $\vec{u} \times \vec{B}$ = the magnetic contribution (it vanishes if the particle is at rest in that frame).

3 Objective

To find how the fields in S' are related to the fields in S:

$$(E'_x, E'_y, E'_z) \longleftrightarrow (E_x, E_y, E_z \;\&\; B_x, B_y, B_z)$$

$$(B'_x, B'_y, B'_z) \longleftrightarrow (E_x, E_y, E_z \;\&\; B_x, B_y, B_z)$$

4 Thought Experiment

A charged particle (charge q) is **instantaneously at rest in the S'-frame**:

$$\vec{u}\,' = 0 \qquad (S' \text{ moving with velocity } \vec{v} \text{ along } x)$$

Force in S'

Since $\vec{u}\,' = 0$, the magnetic term $\vec{u}\,' \times \vec{B}\,'$ vanishes and the Lorentz force in S' is purely electric:

$$F'_x = qE'_x, \qquad F'_y = qE'_y, \qquad F'_z = qE'_z$$

Q: What force will it experience in the S frame?

The particle is at rest in S', and S' moves with velocity v along x relative to S, so in S the particle moves with the frame velocity:

$$u_x = v, \qquad u_y = 0, \qquad u_z = 0 \qquad (\text{in frame } S)$$

Then $\vec{u} \times \vec{B} = (0,\ -vB_z,\ vB_y)$ and $\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$ gives

$$F_x = qE_x, \qquad F_y = q\,(E_y - vB_z), \qquad F_z = q\,(E_z + vB_y)$$

Relating $\vec{F}\,'$ to $\vec{F}$

As the particle is instantaneously at rest in S':

$$F'_y = \gamma F_y, \qquad F'_z = \gamma F_z, \qquad F'_x = F_x$$

Why these follow from the summary formulas (taking $u_x = v$)

With $u_x = v$ we get $\dfrac{\beta u_x}{c} = \dfrac{\beta v}{c} = \beta^2$ and $1 - \beta^2 = \dfrac{1}{\gamma^2}$. Then:

$$F'_y = \frac{F_y}{\gamma(1-\beta^2)} = \frac{F_y}{\gamma \cdot \gamma^{-2}} = \gamma F_y, \qquad F'_z = \gamma F_z$$

$$F'_x = \frac{F_x - \dfrac{\beta}{c}(\vec{F}\cdot\vec{u})}{1-\beta^2} = \frac{F_x - \dfrac{\beta}{c} F_x\, v}{1-\beta^2} = \frac{F_x(1-\beta^2)}{1-\beta^2} = F_x$$

using $\vec{F}\cdot\vec{u} = F_x v$ since only $u_x = v$ is non-zero.

Substituting

Substitute F'_x, F'_y, F'_z (from S') and F_x, F_y, F_z (from S) into these relations. The charge q is the same in both frames and cancels:

$$\begin{aligned} qE'_x &= qE_x && \Longrightarrow && E'_x &= E_x \\ qE'_y &= \gamma\, q\,(E_y - vB_z) && \Longrightarrow && E'_y &= \gamma\,(E_y - vB_z) \\ qE'_z &= \gamma\, q\,(E_z + vB_y) && \Longrightarrow && E'_z &= \gamma\,(E_z + vB_y) \end{aligned}$$

Result: transformation of $\vec{E}$ for a boost along x

$$\boxed{E'_x = E_x}, \qquad \boxed{E'_y = \gamma\,(E_y - vB_z)}, \qquad \boxed{E'_z = \gamma\,(E_z + vB_y)}$$

where: E_x, E_y, E_z and B_y, B_z are field components measured in S; E'_x, E'_y, E'_z are the electric-field components measured in S'; v is the velocity of S' relative to S (along x); $\gamma = 1/\sqrt{1 - v^2/c^2}$.

5 Generalising

Split $\vec{E}$ into the component parallel to the boost direction and the component perpendicular to it:

$$\boxed{E'_\parallel = E_\parallel} \qquad \text{(parallel component)}$$

$$\boxed{\vec{E}'_\perp = \gamma\left(\vec{E}_\perp + (\vec{v}\times\vec{B})_\perp\right)} \qquad \text{(perpendicular component)}$$

where: $\parallel$ and $\perp$ are taken with respect to the direction of $\vec{v}$, the velocity of S' relative to S. (The board writes $\vec{u}$ in the last formula; it is the same vector as $\vec{v}$ here, because a charge at rest in S' moves with velocity $\vec{u} = \vec{v}$ in S.)

Check against the components above: for $\vec{v} = v\hat{x}$, $(\vec{v} \times \vec{B}) = (0, -vB_z, vB_y)$, so $E'_y = \gamma(E_y - vB_z)$ and $E'_z = \gamma(E_z + vB_y)$, exactly as derived.

For reference only: not derived on this page

The objective also lists (B'_x, B'_y, B'_z). The board does not derive it here; the standard result (SI units) for the same boost is

$$B'_x = B_x, \qquad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right), \qquad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right).$$

Treat this as a pointer to the next step of the lecture, not as part of this page's derivation.